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Modeling and Control of a Desktop Robotic Arm for Agricultural Planting

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Abstract

This milestone focuses on the design and simulation of a **Farmer Robotic Arm for Automated Seed Planting**, aimed at automating repetitive agricultural tasks such as seed placement and watering. The analysis was conducted through **mechanical design**, **servo control simulation**, and **software integration** using **Arduino Uno** and **Raspberry Pi** platforms. The system was visualized through 3D CAD modeling and simulated in MATLAB/Simulink to verify motion sequences. Results indicate stable arm movement, accurate positioning of the gripper, and reliable control synchronization between hardware and software modules. The findings demonstrate the potential for efficient small-scale automation in precision agriculture and provide a foundation for future development of fully autonomous robotic farming systems.

Keywords: DH Convention, Robotic Arm, Trajectory Planning, Arduino Control, Raspberry Pi Integration, Agricultural Automation, Mechatronic Design

I. INTRODUCTION

Modern agriculture faces growing challenges due to labor shortages, demand for precision planting, and sustainability concerns. Automation and robotics have become key enablers in addressing these challenges, with robotic arms offering high precision and repeatability for planting and handling operations.

The **Farmer Robotic Arm for Automated Seed Planting** is designed to perform three fundamental agricultural actions:

1. **Seed Planting:** Picking up and placing seeds in soil with controlled precision.
2. **Watering:** Manipulating a hose to water the planted seed through a controlled pump.
3. **Automation Cycle:** Repeating watering operations at preset intervals, managed through programmed logic.

This project integrates **mechatronic design, embedded control, and automation logic** to produce a low-cost, semi-autonomous agricultural robot prototype.

The contribution of this report is to present the design, analysis, and simulation of the robotic arm's motion, mechanical structure, and control logic.

Figure 1 illustrates the conceptual model of the robotic arm.



Fig. 1: The Farmer Robotic Arm

The remainder of this report is organized as follows:

- **Section II** presents the hardware components and circuit design.
- **Section III** introduces the robot's frame assignment and kinematic configuration.
- **Section IV** discusses the DH-convention analysis and modeling approach.
- **Section V** demonstrates the simulation results and visualizations.
- **Section VI** concludes with findings and recommendations for future development.

II. Hardware Components and Circuit Design

To build the robotic arm, several mechanical and electrical components were used to ensure adequate torque, precision, and control flexibility. The system integrates servo motors for actuation, a microcontroller for motion control, and a single-board computer for automation management.

Table 1: Hardware Components Table

Component	Quantity	Place / Source
MG996R Servo Motor 180° 11 kg·cm Metal Gears	3	Ram Electronics
SG90 Micro Servo Motor 180° 1.8 kg·cm Plastic Gears	3	Ram Electronics
Power Supply	1	Futures Electronics
Raspberry Pi	1	Tech Hub
Arduino Uno	1	Ram Electronics
Breadboard	1	Ram Electronics
Jumper Wires Set	–	Ram Electronics
Wires Set	–	–

Component	Quantity	Place / Source
3D Printed Parts	–	–
Screws and Fasteners	–	–
Webcam	1	Amazon
Push Buttons	4	Ram Electronics
Set of Resistors	–	Ram Electronics

For the circuit design we will connect the Raspberry Pi with the Arduino in a circuit connected to a power supply.

III. EX. ROBOT'S FRAME ASSIGNMENT

The robotic arm consists of multiple joints providing **six degrees of freedom (DOF)** to perform precise planting and watering tasks. Each servo motor corresponds to one revolute joint, with coordinate frames assigned following the **Denavit–Hartenberg (DH) convention**.

The reference frame (Base Frame 0) is fixed to the ground, while the following frames (1–6) are assigned to each joint in the order of movement from base rotation to gripper orientation.

This frame assignment facilitates analytical modeling of the arm's kinematics and trajectory control in MATLAB/Simulink.

IV. EX. DH CONVENTION

The robot's forward kinematics were derived using the **DH parameters**, which relate joint rotations to the position and orientation of the end-effector. These parameters define the robotic arm's spatial motion.

Table 2: DH Parameters Table

Link (i)	a_i (mm)	α_i (rad)	d_i (mm)	θ_i (rad)
1	a_1	α_1	d_1	θ_1
2	a_2	α_2	d_2	θ_2
3	a_3	α_3	d_3	θ_3
4	a_4	α_4	d_4	θ_4
5	a_5	α_5	d_5	θ_5
6	a_6	α_6	d_6	θ_6

The DH table enables systematic analysis of the end-effector's motion, providing the basis for simulation and control algorithms.

V. SIMULATION RESULTS

In the Simscape Multibody simulation, the mechanical model of the robotic arm was first imported from SolidWorks into MATLAB. Initially, the motors caused continuous, unrestricted motion because the system lacked physical constraints — there was no ground reference or frictional contact to limit rotation.

To resolve this issue, a ground reference was introduced and contact force blocks were applied between the ground and the arm's base components. This allowed the system to realistically simulate the interaction between the robot and the floor, constraining unwanted motion and stabilizing the structure.

A comparison between the two configurations is shown in Figure X and Figure Y — the first illustrating the free-moving, unconstrained model, and the second showing the stabilized model after implementing the ground and contact forces.

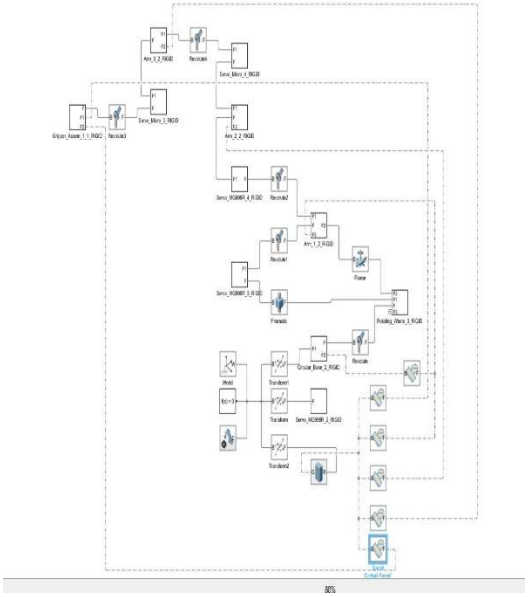


Fig. 2: Simulink Simulation

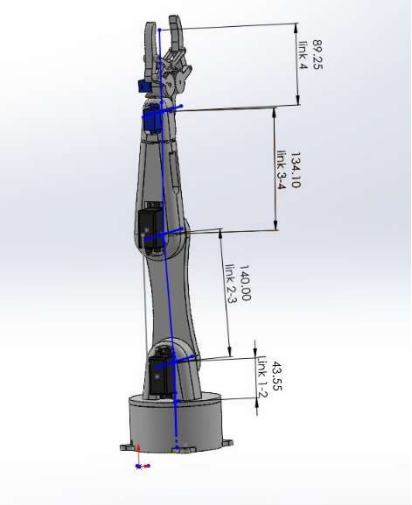


Fig. 3: The Farmer Robotic Arm Parameters

VI. CONCLUSINS AND FUTURE RECOMMENDATIONS

This milestone achieved the preliminary design, modeling, and simulation of the **Farmer Robotic Arm for Automated Seed Planting**. The system successfully integrates you are going to implement in the upcoming milestone. mechanical, electrical, and software components into a functioning prototype.

Conclusions:

- The robotic arm can execute precise planting and watering tasks.
- Integration between Arduino (low-level control) and Raspberry Pi (high-level control) was effective.
- Simulation verified the robot's stability and functional sequence.

Future Recommendations:

- Implement **vision-based control** for seed and soil detection.
- Integrate **IoT-based remote monitoring** for large-scale field deployment.
- Enhance **path planning** through inverse kinematics and feedback optimization.

These improvements will enable full autonomy and real-world agricultural applicability in future milestones.

VII. APPENDIX

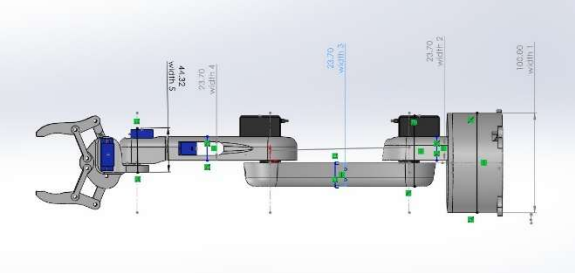


Fig. 4: The Farmer Robotic Arm Parameters

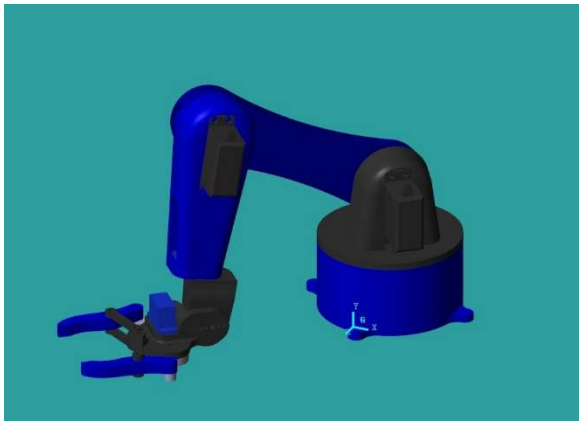


Fig. 5: The Farmer Robotic Arm 3D